

ABIGAEK KALUNDE SILA



Graduate Electrical and Electronics Engineer with practical experience across utility, transportation, aviation, and data center power infrastructure. Experienced in high-voltage substations, electrical distribution systems, backup power solutions, SCADA monitoring, and energy performance analysis. Strong foundation in power system design principles, electrical network planning, protection systems, and resilient infrastructure development. Passionate about designing reliable, efficient, and sustainable power systems that support critical facilities and next-generation digital infrastructure.

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🌐 abigael-sila.github.io/Abigael-Portfolio-Website

KEY AREAS OF COMPETENCE

Power Systems & Electrical Infrastructure:

- Exposure to high-voltage and low-voltage power systems, transformers, switchgear, and electrical distribution systems; diesel generators, UPS, and backup power systems; SCADA and DCIM monitoring and substation operations; electrical fault identification and troubleshooting; single-line diagram and control schematic interpretation; energy efficiency and power reliability fundamentals; Airfield Ground Lighting (AGL) and Constant Current Regulator (CCR) operations.

Testing, Safety & Technical Practices

- Exposure to the use of electrical testing equipment including insulation resistance testers, clamp meters, multimeters, earth resistance testers, and power quality analyzers, Lock-Out/Tag-Out (LOTO) procedures, equipment inspection and preventive maintenance, compliance with industrial and aviation safety requirements

Operations & Professional Skills

- Exposure to technical report writing and maintenance documentation; equipment data collection and analysis; maintenance planning support; engineering drawing, schematic, and single-line diagram interpretation; problem-solving and root cause investigation; preparation of inspection records, maintenance logs, procurement and inventory support, cross-departmental collaboration.

Electronics & IoT

- Exposure to PCB design and prototyping; embedded systems development using ESP32, Arduino, and A9G platforms; sensor integration and data acquisition; wireless communication technologies including Wi-Fi and GPRS.

WORK EXPERIENCE

Data Center Intern Safaricom PLC

05/2026 - Present

- Supporting the operation and maintenance of critical data center infrastructure under the guidance of experienced engineers.
- Learning the design and planning principles of data center electrical and mechanical systems through on-the-job training, technical discussions, and assigned research tasks.
- Studying power distribution systems and tracing electrical single-line diagrams from utility supply through transformers, switchgear, UPS systems, and critical IT loads.
- Assisting with monitoring of data center systems using DCIM and other monitoring platforms to observe equipment performance, alarms, and operational trends.
- Participating in routine inspections of electrical, mechanical, and support infrastructure to improve understanding of equipment operation and reliability requirements.
- Supporting the review and interpretation of engineering drawings, layouts, and technical documentation used in data center projects.
- Conducting technical research on data center equipment and infrastructure components and presenting findings to engineering teams.
- Learning to identify infrastructure gaps, risks, and improvement opportunities and contributing observations during engineering discussions.
- Assisting in the preparation and updating of technical documentation, asset records, and operational reports.
- Collaborating with engineers across operations, maintenance, planning, and design functions to gain exposure to the full lifecycle of data center infrastructure projects.

Electrical Engineering Attachee Kenya Airports Authority - JKIA

08/2025 - 12/2025

- Supported maintenance of terminal and substation electrical systems, including 66kV and 11kV switchgear, transformers, airfield lighting circuits, capacitor banks, rectifiers and voltage regulators while strictly following aviation safety procedures.
- Assisted in upkeep of backup power systems, including 2 MVA diesel generators, UPS units, battery inverters and automatic/manual changeover systems, ensuring reliable standby power and operational safety compliance.
- Contributed to maintenance of airfield ground lighting, runway and taxiway circuits, signage and operation of Constant Current Regulators, enhancing airfield lighting reliability.
- Gained hands-on exposure to aviation systems, including Airport Lighting System, Instrument Landing System, radar and air traffic control support systems, through supervised technical tasks and briefings.

- Participated in energy performance assessments, providing observations used to optimize power consumption and improve energy efficiency across terminals and substations.
- Developed practical experience interpreting single-line diagrams and schematics for maintenance planning, system tracing, and equipment identification.
- Developed competence in electrical testing and diagnostic tools, including insulation resistance testers, multimeters, clamp meters and

Electrical Engineering Attachee

Kenya Space Agency

08/2025

- Gained foundational knowledge of the Kenya Space Agency's role in regulating Kenya's air and space domain, including satellites, rockets and shuttles under international collaboration.
- Observed the design lifecycle of mission-critical aerospace electronics, including PCB design for embedded systems and IoT applications in satellite and rocket systems.
- Acquired exposure to aerospace power system design principles, including battery backup solutions for rockets and power infrastructure for ground stations.

Electrical Engineering Attachee

Kenya Railways Corporation – SGR Nairobi

05/2024 - 12/2024

- Assisted in inspection, cleaning, testing, and minor repairs of 66kV, 33kV, and 11kV transformers, unmanned stations, box-type transformers, switchgear, and distribution boxes, ensuring operational safety and equipment reliability.
- Conducted inspections and handled manual and automatic 75 kVA diesel generators, monitoring ventilation, oil and coolant levels, filters, belts, leaks, startup behavior, voltage, current, frequency, and shutdown performance.
- Carried out field inspections of overhead power lines using linking sticks, telescopic pruning tools, insulated cutters, climbing gear, and grounding equipment, ensuring hazard clearance and structural stability.
- Supported real-time monitoring of substations via SCADA, observing voltage, current, alarms, and load trends, and participated in switching operations to maintain system stability.
- Assisted in executing switching and coupling procedures during power line failures, including circuit breaker resetting and restoration of supply, minimizing downtime.
- Helped troubleshoot and repair building electrical systems, including sockets, internal wiring, floodlights, tower lights, street lamps, and distribution panels, resolving faults such as tripped breakers, burnt plugs, loose connections, and damaged cables.
- Recorded daily power consumption and supported generator performance assessments for energy-efficiency reviews across SGR operations and third-party users.
- Assisted in procurement, handling, and verification of generator oil, batteries, fittings, and tools, ensuring correct specifications and smooth logistics for maintenance activities.
- Completed inspection logs, maintenance records, and switching notes, while supporting administrative tasks such as financial form sorting and tender file organization.
- Collaborated with engineers and technicians during field operations, guided small tasks, assisted newer team members, and contributed to seamless coordination and execution of maintenance activities.

KEY ACHIEVEMENTS

- Designed, prototyped, and tested Symphion, an IoT-based emergency response device integrating GPS tracking, GSM/GPRS communication, cloud connectivity, and custom PCB design.
- Gained practical exposure to high-voltage substations, electrical distribution systems, SCADA operations, and backup power infrastructure across railway, aviation, and data center environments.
- Supported operation and maintenance activities that contributed to the reliability and continuity of critical infrastructure systems.
- Developed hands-on experience in electrical inspections, testing, troubleshooting, technical documentation, and infrastructure monitoring.

EDUCATION

BEng Electrical and Electronics Engineering

Moi University

01/2021 - 07/2025

- **Capstone Project:** Designed and developed Symphion, an Embedded Systems & IoT prototype SOS device with GPS tracking, live audio, GSM/GPRS connectivity, and custom PCB design.

Kenya Certificate of Secondary Education

Asumbi Girls' High School

02/2016 - 11/2019

SOFTWARE TOOLS

- SCADA Systems and DCIM Platforms
 - AutoCAD
 - EasyEDA
- Android Studio
 - Visual Studio Code
 - Git & Github

CERTIFICATIONS

Meta Front-End Developer Professional Certificate

A 9-course professional certification from Meta focused on end-to-end front-end development.

July 2024

Meta Android Developer Professional Certificate

A 12-course professional certification from Meta focused on end-to-end Android mobile application development.

December 2025

Business Modelling for Entrepreneurs

Open University of Kenya Professional Development Certificate focused on business model design, analysis, and innovation.

December 2025

SOFT SKILLS

- Charismatic
 - Proactive
 - Flexible
 - Agile
- Adaptable
 - Collaborative
 - Self-motivated
 - Effective communicator

REFERENCES

ESTHER WANGARI KARUGA, Senior Manager Data Center Lead, Safaricom PLC.

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ENG. ZABLON ONTWEKA, Airport Engineer, Kenya Airports Authority.

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AMOS WANJOHI MWANGI, Senior Electrical Technician, Kenya Railways – SGR Nairobi.

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